

Wireless synchronization of Trinet cameras — validation report

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Summary

Trinet cameras in a **Wrist Kit** have no cables between them; each keeps time with its own oscillator, and a short-range radio link brings all of them onto one shared clock. This report measures how closely the resulting timestamps agree, using an **optical reference that is independent of every clock under test**.

Result	Value
Camera-to-camera offset, wrist vs stereo head (median, ~7,000 frames)	+56 μs
Mean offset over 27 sessions, each after a full cold restart	+49 $\pm$ 8 μs (mean $\pm$ standard error)
Session-to-session variation (standard deviation)	43 μs
Drift over one continuous hour	None measurable (+0.3 $\pm$ 5.5 μ s per hour)
Stereo head camera, left vs right eye	0.0 μs median, 0.82 μ s standard deviation

All measured offsets are a small fraction of the 33.3 ms frame interval at 30 fps. We specify camera-to-camera agreement conservatively as **under 1 ms**.

1. Test setup

Cameras	A Trinet Stereo (rolling-shutter) head camera and Trinet wrist cameras, paired as a kit; a two-camera kit for the one-hour test
Link	The cameras' built-in radio link, operating live during recording
Recording	1920 $\times$ 1080 at 30 fps to each camera's own memory card
Firmware	August 2026 builds
Environment	Indoor bench; exposure locked at 1/120 s for the one-hour test

2. Method

Optical ground truth. An LED switching at randomized times was placed in view of every camera. Because a rolling-shutter sensor exposes its rows one after another (about 29 μ s per row), a frame captured while the LED switches shows a brightness edge part-way down the image; the edge's row position locates the switching instant within the frame to a fraction of a row. Comparing that instant with each camera's own frame timestamps gives the timestamp error of every camera against the same physical event — without reference to any clock being tested, and without per-unit calibration inputs.

Frame pairing. For the one-hour test, frames from the two cameras were paired by their timestamps (precision about 8 μ s) to measure frame-to-frame variation and long-term drift. Pairing measures variation and drift; the optical method measures the absolute offset.

Exclusions. The first second of each recording, while a camera locks onto the shared clock, is excluded from the statistics and shown shaded in Figure 1.

3. Results

3.1 Per-frame agreement

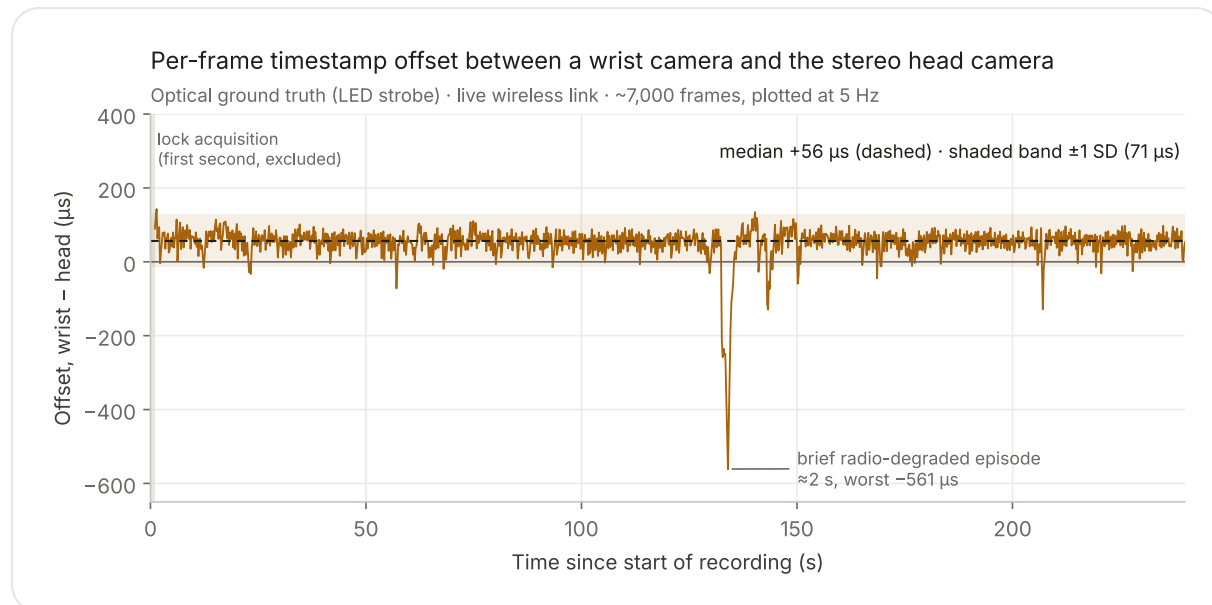


Figure 1. Timestamp offset between a wrist camera and the stereo head camera for light-matched frames over a four-minute recording (~7,000 frames, plotted at 5 Hz). Median +56 μ s, standard deviation 71 μ s. Over 99% of plotted samples lie within 150 μ s of zero. One brief radio-degraded episode, about 2 s long, reached -561 μ s before the link recovered.

3.2 Repeatability across cold restarts

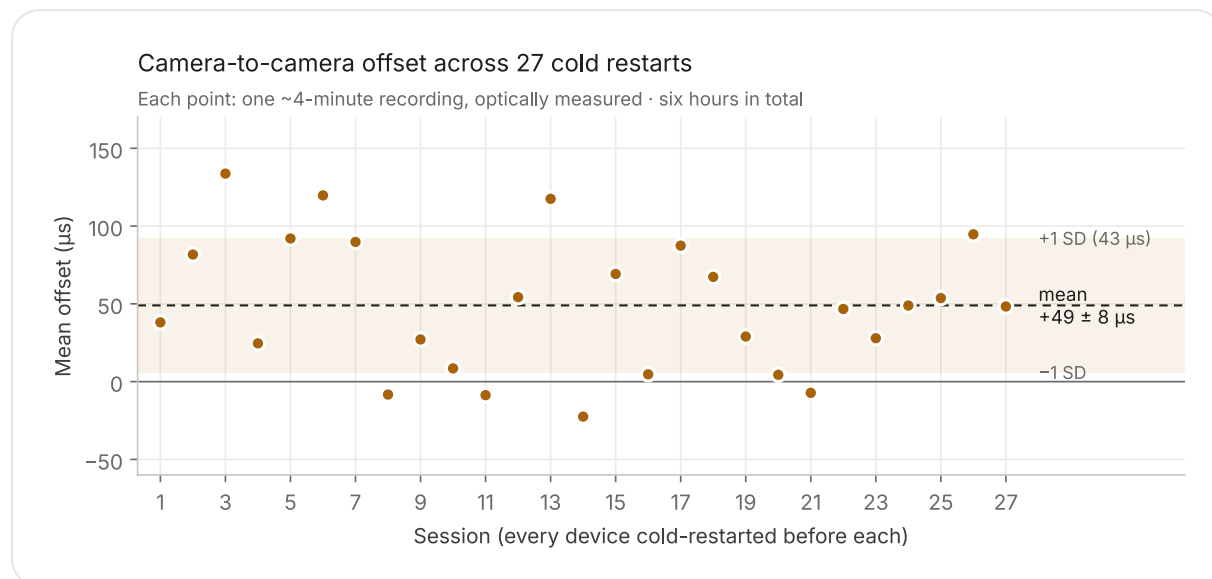


Figure 2. Mean offset in each of 27 sessions (about four minutes each, six hours in total), with every device fully restarted before each session — the worst case for any state carried between recordings. Mean +49 $\pm$ 8 μ s (standard error), standard deviation 43 μ s, range -22 μ s to +134 μ s. The shaded band is ± 1 standard deviation; no trend across sessions.

3.3 Stereo head camera: left vs right eye

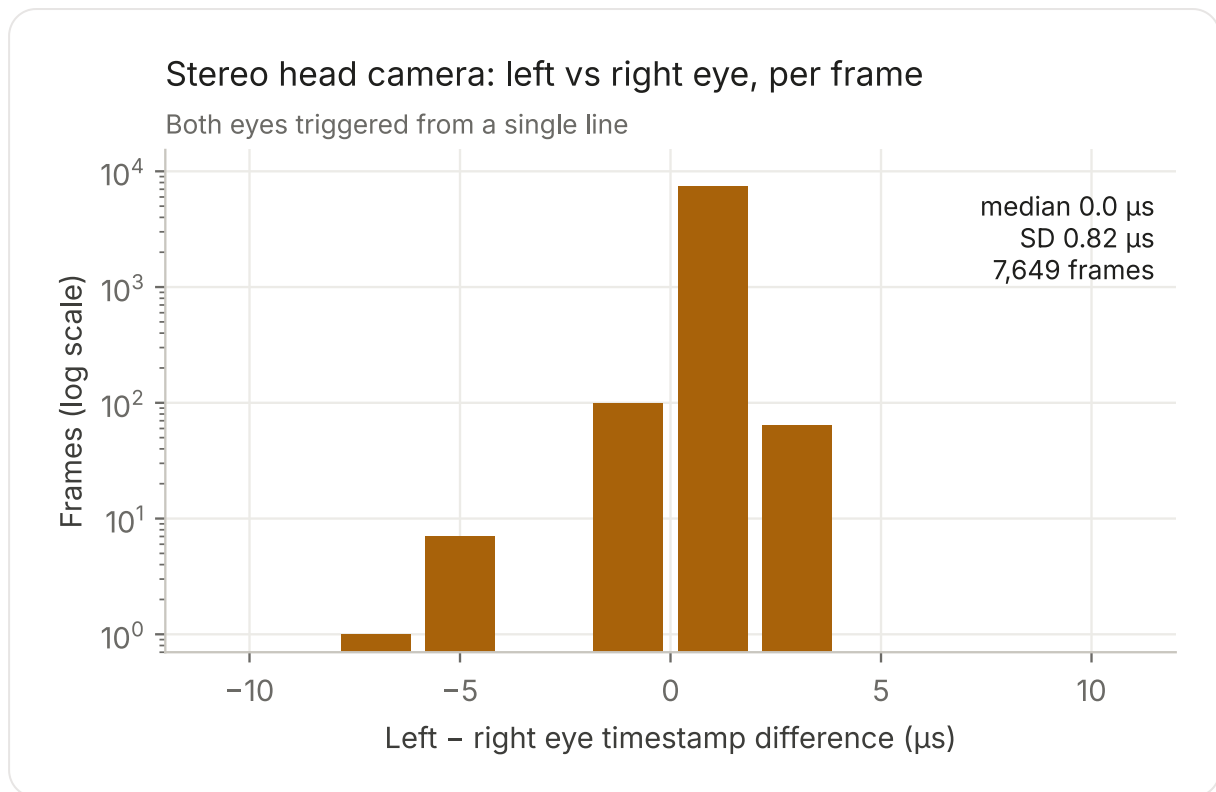


Figure 3. Per-frame timestamp difference between the two eyes of the stereo head camera over a full recording (7,649 frames; logarithmic scale). Median 0.0 μs , standard deviation 0.82 μs ; 99.1% of frames within $\pm 2 \mu\text{s}$. Both eyes are triggered from a single line, so this also bounds the error of the timestamping chain itself at under a microsecond.

3.4 Long-duration stability

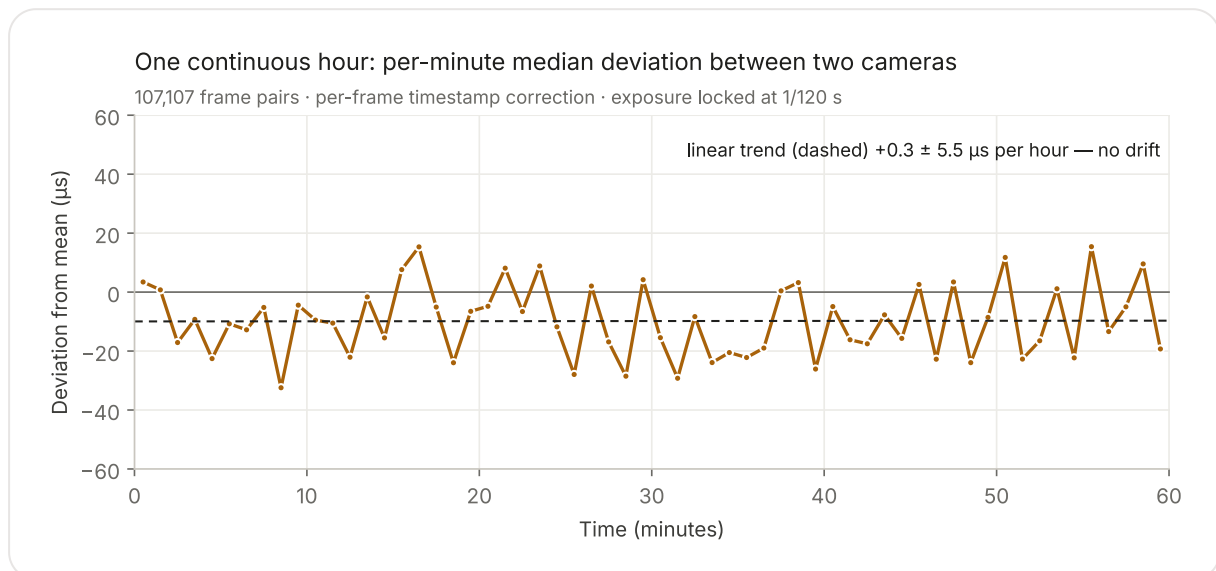


Figure 4. Per-minute median deviation from the session mean between two kit cameras over one continuous hour (107,107 frame pairs). Linear trend +0.3 $\pm$ 5.5 μs per hour; per-frame trend $-1.0 \pm 0.7 \mu\text{s}$ per hour. Over the same hour the cameras' raw clocks drifted apart by 14.4 ms; per-frame timestamp correction removes this to within about 1 μs .

Frame-to-frame deviation, one hour	Value
Median (p50)	33.7 μ s
90th percentile	86.6 μ s
99th percentile	230 μ s
Standard deviation	66 μ s

4. Specification

Parameter	Typical value	Basis
Camera-to-camera offset	$\approx 50 \mu$ s	Figures 1 and 2
Session-to-session variation	$\approx 43 \mu$ s (1 SD)	Figure 2
Frame-to-frame deviation, 99th percentile	$\approx 230 \mu$ s	Section 3.4
Drift over one hour	Not measurable ($\leq 1 \mu$ s)	Figure 4
Stereo left vs right eye	$< 1 \mu$ s	Figure 3
Product specification, camera to camera	Under 1 ms	All of the above, with headroom

5. Limitations

- **Lock acquisition.** During the first one to two seconds of a recording, while a camera locks onto the shared clock, timestamps can be less accurate. Trim this period where the tightest alignment matters.
- **Radio-degraded episodes.** Brief interruptions of the radio link degrade agreement temporarily: $\sim 561 \mu$ s at worst in the four-minute test. In the one-hour test, three re-lock episodes of about 0.5 s produced errors of up to about 11 ms. Each frame carries a quality flag that marks most such frames; use it to exclude them.
- **Conditions.** Results are from indoor bench tests with two or three cameras at short range. Results in the field depend on the environment and the radio conditions.

6. Data availability

Methodology details and raw measurement data are available on request — [contact us](#). Related reading: [Timing and sync](#) · [Wrist Kit](#) · [Sample data](#).